

CUMSHEWA ISLAND, BC, Canada

WMO: 717710

Lat: 53.030N

Lon: 131.601W

Elev: 46

StdP: 14.67

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	26.8	30.0	14.6	11.5	29.7	19.4	14.6	31.2	50.5	41.5	46.4	43.3	13.1	330	0.723

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.3	65.8	59.3	63.6	58.2	61.9	57.3	60.6	64.3	59.4	62.8	58.4	61.5	11.3	350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
58.8	74.1	62.2	57.8	71.4	61.0	56.9	69.2	60.0	26.8	64.5	26.0	63.1	25.4	61.7	65.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 42.8	36.6	31.3	DB	23.7	70.8	6.4	2.6	19.1	72.7	15.3	74.2	11.8	75.7	7.1	77.6	(4)
(5)			WB	21.2	62.6	7.2	1.3	16.0	63.5	11.8	64.3	7.7	65.0	2.4	66.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	48.2	41.7	40.9	41.3	44.4	48.4	52.8	57.3	58.7	55.5	50.0	44.6	42.0		
	DBStd	7.43	5.40	4.32	4.27	3.00	3.68	3.76	3.07	2.67	2.70	3.30	4.47	4.68		
	HDD50	1490	259	256	269	170	72	9	0	0	0	39	167	249		
	HDD65	6141	723	676	734	618	514	365	239	197	284	464	613	714		
	CDD50	827	0	0	0	2	23	95	227	269	167	40	4	0		
	CDD65	3	0	0	0	0	0	1	1	1	0	0	0	0		
	CDH74	2	0	0	0	0	0	2	0	0	0	0	0	0		
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0		
	WSAvg	12.5	15.7	14.3	14.0	13.2	11.4	10.3	8.9	9.2	10.7	12.7	14.5	15.4		
	PrecAvg	89.1	10.9	8.0	7.7	5.7	4.1	2.9	2.4	3.6	5.8	12.2	11.9	12.5		
(15) Precipitation	PrecMax	106.9	18.5	15.5	11.8	8.7	9.4	6.1	5.3	9.0	12.3	19.4	21.3	26.5		
	PrecMin	76.5	5.6	3.0	4.6	2.2	1.9	1.2	0.9	1.3	1.9	6.0	6.3	7.8		
	PrecStd	8.6	3.1	3.1	2.2	1.5	1.9	1.3	1.2	1.9	2.2	3.3	3.3	4.1		
	PrecStd	8.6	3.1	3.1	2.2	1.5	1.9	1.3	1.2	1.9	2.2	3.3	3.3	4.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.3	50.5	51.2	55.1	60.1	66.4	69.5	69.1	64.3	60.1	54.8	51.7		
		MCWB	48.4	47.5	47.2	46.7	51.8	56.8	59.9	60.4	57.4	55.2	50.5	48.2		
	2%	DB	50.2	48.8	49.1	51.5	56.8	61.4	65.3	66.3	61.9	57.1	52.4	50.0		
		MCWB	47.6	46.4	45.8	45.9	51.1	55.7	59.4	59.5	56.6	53.4	49.4	47.3		
	5%	DB	48.9	47.6	47.9	49.6	54.6	59.1	63.0	64.1	60.3	55.3	51.2	48.9		
		MCWB	46.8	45.5	45.0	45.7	50.0	55.1	57.9	58.9	55.9	51.9	48.7	46.6		
	10%	DB	47.9	46.5	46.8	48.3	52.9	57.3	61.1	62.4	59.1	54.0	50.2	47.9		
		MCWB	46.0	44.4	44.1	45.1	49.1	53.9	56.9	58.1	55.3	51.0	47.9	45.8		
	0.4%	WB	49.4	49.0	48.5	49.0	54.1	58.8	62.0	62.3	59.9	56.1	51.6	49.1		
		MCDB	50.6	50.2	50.3	50.3	57.8	62.5	66.3	67.2	63.2	58.3	52.5	50.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	48.1	47.3	46.7	47.7	52.2	57.2	60.3	61.0	58.2	54.5	50.1	47.8		
		MCDB	49.3	48.4	48.6	49.4	55.5	60.6	64.2	64.8	60.6	56.3	51.2	49.0		
	5%	WB	47.1	46.0	45.8	46.7	51.0	55.8	59.1	60.0	57.2	53.1	49.1	47.0		
		MCDB	48.4	47.4	47.6	48.6	53.6	58.8	62.5	63.5	59.3	54.7	50.6	48.4		
	10%	WB	46.1	44.9	44.7	45.7	50.0	54.3	57.9	59.0	56.3	51.9	48.0	46.0		
		MCDB	47.5	46.4	46.5	47.8	52.1	56.9	60.9	62.3	58.5	53.6	49.9	47.6		
	5% DB	MDBR	5.3	5.4	5.7	5.5	5.4	5.2	5.3	5.3	5.5	5.6	5.7	5.6		
		MCDBR	5.6	5.7	6.2	6.7	7.8	8.2	8.1	7.9	6.9	6.8	6.0	5.9		
		MCWBR	4.8	5.3	5.1	4.9	5.0	5.1	4.6	4.9	4.9	5.4	5.6	5.2		
		MCWBR	5.4	5.5	6.1	5.6	6.6	7.2	7.6	7.2	5.9	6.4	5.9	5.6		
(35) Mean Daily Temperature Range	5% WB	MCWBR	4.9	5.2	5.3	4.8	4.5	4.7	4.5	4.9	4.8	5.5	5.6	5.3		
		MCWBR	4.9	5.2	5.3	4.8	4.5	4.7	4.5	4.9	4.8	5.5	5.6	5.3		
		MCWBR	4.9	5.2	5.3	4.8	4.5	4.7	4.5	4.9	4.8	5.5	5.6	5.3		
		MCWBR	4.9	5.2	5.3	4.8	4.5	4.7	4.5	4.9	4.8	5.5	5.6	5.3		
(40) Clear-Sky Solar Irradiance	taub		0.319	0.325	0.350	0.388	0.386	0.373	0.374	0.374	0.368	0.351	0.328	0.309		
	taud		2.372	2.464	2.407	2.286	2.357	2.447	2.487	2.500	2.476	2.417	2.325	2.323		
	Ebn at Noon		209	250	266	267	274	279	276	270	257	234	200	187		
	Edh at Noon		19	23	29	37	36	34	32	30	27	23	19	16		
(44) All-Sky Solar Radiation	RadAvg		186	455	805	1265	1601	1675	1600	1353	934	522	249	154		
	RadStd		22	72	91	63	127	125	155	111	92	61	26	21		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+135	N/A
(47)	Regional (16 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page